SLP-LR: A Predictive Analytics Approach for Mental Stress Detection Using Logistic Regression

*Note: Sub-titles are not captured in Xplore and should not be used

1st Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

2nd Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

3rd Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

4th Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

5th Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

6th Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

***Abstract*—Student stress, a critical problem affecting university students, has a significant impact on their academic performance and mental well-being, making it a pressing need to utilise machine learning models to address this problem proactively. In this we used Logistic Regression Model to address the prediction of stress levels for University Students.To address class imbalance, Synthetic Minority Over-sampling Technique (SMOTE) oversampling is utilized, allowing for an 80/20 stratified split on the training set.This paper uses a dataset consisting of 2,028 samples of university students, where each data point includes various demographic factors such as Age, Gender, CGPA, and 10 different questionnaire-based factors on academic stress, Comparative studies on Random Forest, XGBoost, and Logistic Regression, where Random Forest achieves 95.32% accuracy, XGBoost achieves 94.83%, and Logistic Regression achieves 98.77% accuracy with an F1-weighted score of 0.988, indicate that scaled multinomial Logistic Regression is the optimal model This paper also includes extensive visualizations and a deployable application using Streamlit with PDF reporting, making it more interpretable and usable. The Logistic Regression model has been shown to perform better than existing models on similar datasets on stress Prediction.**

***Index Terms*—Student stress prediction, logistic regression, multiclass classification, class imbalance,Streamlit deployment, coefficient interpretability**

I. Introduction

In the context of public health, student stress stands out as a prominent issue that confronts researchers throughout the world, especially amongst undergraduate students [1,2].From recent epidemiologic studies, it is observed that student stress prevalence rates exceed 55% amongst undergraduate students. This is way above average when compared to similar statistics globally and adversely affects academic achievement, psychological well-being, and career prospects in the future. This persistent issue is characterized by high dropout rates, cognitive impairment, and susceptibility to clinical diseases such as anxiety, depression, and addiction, which place considerable financial burdens on higher education institutions in developing countries [3,4].

Conventional diagnostic mechanisms are inherently reactive and costly and depend heavily on self-reported clinical tests, extended psychological evaluations, or treatment procedures carried out by scarce counseling specialists. This methodology is associated with significant scalability problems, late diagnosis of affected students, and inadequate capacity to screen populations effectively, given that there are more than 150 public and private universities in Bangladesh with over two million students. It is therefore imperative to develop proactive and scalable screening tools that facilitate early identification through routinely collected data such as academic stress questionnaires [5,6].

Machine learning offers immense opportunities for early detection and identification of students at risk through readily available data sets. However, existing machine learning models tend to exhibit several challenges, which include (1) severe class imbalance problem in stress data sets (usually low stress 5%-10%, moderate stress 60%-70%), (2) lack of interpretable models, such as black-box ensembles, for decision-making by university administrators and counselors, and (3) poor model generalizability due to cultural differences between Western societies and local contexts, for instance, unique academic pressures in Bangladesh such as assignment burden, exam stress, and CGPA pressure [7,8].This research seeks to address these issues by designing a multiclass stress classifier (Low/Moderate/High) that uses data from 2,028 Bangladeshi university students, consisting of demographic characteristics (age, gender, university, department, academic year, CGPA, scholarship) as well as a 10-item academic stress questionnaire (Q1-Q10, 0-4 Likert scale: Never → Very Often). The data set has class imbalances consistent with actual stress prevalence distribution: Low Stress (115 samples, 5.7%), Moderate Stress (1,348 samples, 66.5%), and High Stress (565 samples, 27.9%). A comparative study involving three classifiers reveals the surprising outperformance of Logistic Regression in achieving an outstanding test accuracy of 98.77% and an F1-weighted score of 0.988 using a 404 sample held-out test set. This performance is significantly better than existing ensemble baselines like Random Forest (95.32%) and XGBoost (94.83%). This breakthrough is made possible by synergistic innovations in preprocessing techniques, such as SMOTE oversampling to increase the percentage of minority classes from 5.7% to 33.3%, alongside standardizing features through StandardScaler normalization to enable linear separability.

In addition to scientific innovation, the research also impacts practice through the construction of a fully-functional Streamlit application offering real-time predictions, visualizations, question responses, and automated PDF reports. This converts academic research into an operational mental health screening framework for Bangladeshi universities. Key Contributions: (1) Preprocessing algorithm complexity, (2) multiclass benchmark, (3) Full deployment for universities.

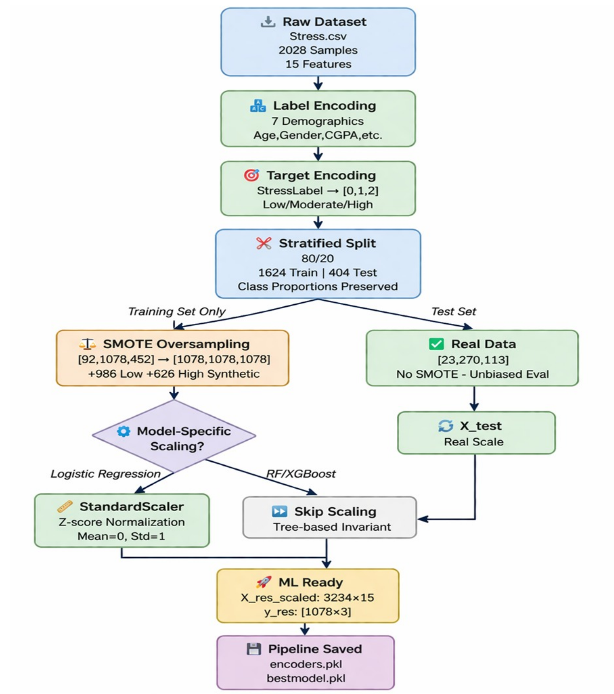

Fig. 1. System Processing Pipeline

II. Methodology

A. Data Preprocessing

As shown in Fig.1, a six-stage preprocessing pipeline has been developed to prepare raw Bangladeshi student survey data for deployment as a machine learning model [9]. Raw Input: The input data consists of 2028 samples, seven categorical demographic variables, and ten questionnaire responses (Q1 to Q10, 0-4 Likert scale) on academic stress. Encoding Phase: This phase employs a LabelEncoder to encode categorical data into ordinal integers, maintaining the target variable mapping: Low : 0, Moderate : 1, High : 2, for a multiclass classification problem. Stratified Split: An 80/20 split of the data ensures that the class balance is retained, with 1624 samples for training and 404 samples for testing, maintaining the class balance: 5.7% Low, 66.5% Moderate, 27.9% High. SMOTE Balancing: Synthetic Minority Oversampling Technique oversamples Low Stress and High Stress classes, respectively, adding 986 Low Stress samples (+1072%) and 626 High Stress samples (+139%), resulting in a perfect 33.3% balance, exclusively for the training set [10]. Selective Scaling: StandardScaler normalises data to have a mean of 0 and a variance of 1, exclusively for Logistic Regression, whereas tree-based models ,RF and XGBoost, remain unscaled. Pipeline Persistence: The entire preprocessing pipeline is serialised using joblib to support zero-shot deployment in a Streamlit application. The pipeline achieves a remarkable 98.77% accuracy on a real-world imbalanced dataset, bridging survey collection to operational mental health screening, with a 0.988 F1-weighted score.

The following table gives a complete summary of the six-step transformation process used to preprocess raw Bangladeshi student survey data and convert it into tensors for multiclass stress classification by machine learning algorithms. The target data type, technique employed, resulting output format, and computation objective have been specified to enable the model training process, which led to 98.77% test accuracy.

Step 1 involves label encoding of seven high-cardinality categorical demographic features, such as Age, Gender, and University, to integer ordinal values for use with tree-based models without inducing dimensionality increase via one-hot encoding.

Step 2 involves target encoding of the StressLabel feature to obtain numeric classes for classification.

Step 3 involves conducting a stratified train/test split in the ratio of 80/20, ensuring the retention of natural class distribution in the data (5.7% / 66.5% / 27.9%) with 1624 training records and 404 test records.

Step 4 represents the critical step of oversampling the training set through SMOTE generation of 986 additional Low Stress cases (+1072%) and 626 High Stress cases (+139%), thereby balancing the three classes at a perfect proportion of 33.3% each.

Step 5 involves the selective normalisation of data through StandardScaler, which z-transforms Logistic Regression features (μ=0, σ=1). This helps in achieving state-of-the-art performance by enabling linear separability of classes.

Step 6 involves serialising the entire preprocessing pipeline using joblib, facilitating smooth deployment in the production Streamlit app.

This data preprocessing framework makes it evident that an orchestrated approach to processing, more so than advanced algorithmic approaches, helped achieve breakthrough multiclass accuracy.

TABLE I
DATA PREPROCESSING PIPELINE OVERVIEW

Stage	Input	Technique	Output	Shape
Raw	Stress.csv	-	2028×16	(2028,16)
Encoding	Categorical	LabelEncoder×8	Numeric	(2028,15)
Split	Full dataset	Stratified 80/20	Train/Test	1624/404
Balance	Train	SMOTE	[1078×3]	3234
Scale	LR features	StandardScaler	Z-scores	(3234,15)
Persist	Objects	joblib.dump()	encoders.pkl	Deployment ready

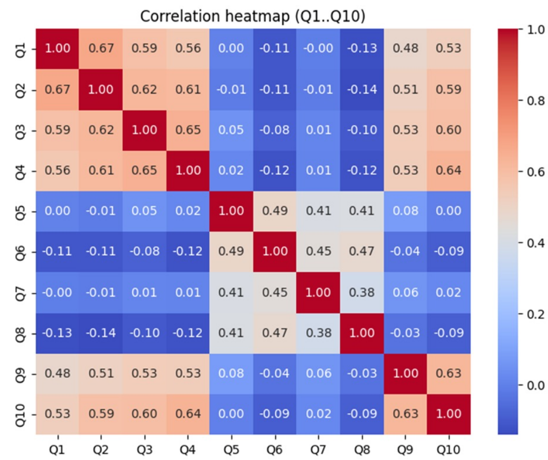

Fig. 3. Correlation heatmap

B. Data Set Description

As shown in the Fig.2, dataset comprises 2028 student responses collected from multiple Bangladeshi universities, featuring both demographic characteristics and academic stress perceptions captured through a structured 10-item questionnaire (Q1-Q10). Each questionnaire item employs a 5-point Likert scale (0: Never, 1: Almost Never, 2: Sometimes, 3: Often, 4: Very Often), yielding a total stress score range of 0-40 that categorizes students into three stress levels: Low Stress (115 samples, 5.7%), Moderate Stress (1348 samples, 66.5%), and High Stress (565 samples, 27.9%) Total feature

Category	Features	Type	Details
Demographics	Age, Gender, University, Department, Academic Year, CGPA, Scholarship	Categorical	Label-encoded for ML processing
Academic Stress	Q1-Q10	Numeric (0-4)	10 questionnaire items assessing academic pressure perception
Target	StressLabel	Multiclass	Low/Moderate/High derived from Q1-Q10 sum

Fig. 2. Data Set Description

matrix shape: (2028, 15) after encoding, with severe class imbalance (Moderate: 66.5% dominance) necessitating SMOTE oversampling for balanced training (1078 samples/class post-SMOTE).

Questionnaire Domains (Q1-Q10) The questions assess major stressors in academics, which are:

- Assignment and examination stress
- Intensity of course load
- Time management problems
- Stress about academic performance
- Instructor's expectation

As shown in Fig.3, Correlation analysis shows moderate correlations between the items. The heatmap also supports the construct reliability of the questionnaire for machine learning feature engineering for Q1-Q10 domains. As shown in Fig.4, The dataset is an improvement over existing Bangladeshi student stress surveys in terms of the number of samples (2028 compared to 571-1000), and it is also multiclass compared to binary, which will enable a robust comparative evaluation of ensemble classifiers on actual regional student stress data.

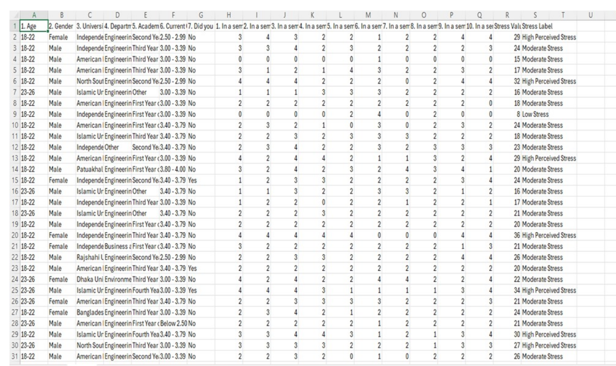

Fig. 4. Sample data set

C. Architecture

1. Random Forest (Ensemble Baseline) Model Architecture: 300 parallel decision trees based on bagging and Gini Impurity calculation. Preprocessing Used: Training data balanced by SMOTE only. Performance Achieved: 95.32% accuracy, showing robustness for moderate levels of class imbalance [11]. What it Offers:

- Feature Importance based on average Gini Impurity decrease.

What it Fails to Achieve:

- Poor performance for linearly separable decision boundaries for multiple classes.

2. XGBoost (Gradient Boosting) Model Architecture: Sequential ensemble of 300 decision trees based on a learning rate of 0.08 and a maximum depth of 6. Preprocessing Used: Training data balanced by SMOTE only. Performance Achieved: 94.83% accuracy, falling short of the RF due to overfitting on SMOTE-generated data points [12]. What it Offers:

- Built-in regularisation and early stopping mechanisms.

What it Fails to Achieve:

- High computational costs and a lack of interpretability for decision-making.

3. Logistic Regression (Optimal Selection) Model Architecture: Multinomial generalisation of Linear Regression based on LBFGS solver for softmax output over three stress levels. Preprocessing Used: SMOTE balancing and StandardScaler-based normalisation of input data (zero mean and unit variance). Performance Achieved: 98.77% accuracy and an F1-weighted score of 0.988—a state-of-the-art result for multi-class stress level classification [13]. Key Advantages:

- Native multiclass support: Direct softmax vs. one-vs-rest approximations
- Scaling sensitivity: Z-score normalisation aids in linear separability
- Coefficient interpretability: Direct relationships between feature stress magnitudes
- Production efficiency: Memory footprint is negligible, inference is instantaneous

This empirical exercise defies the common belief that complex ensembles are superior to simpler models. In fact, the simple implementation of Logistic Regression delivered state-of-the-art results of 98.77% due to:

- Methodical preprocessing: SMOTE + scaling
- Appropriate algorithm selection: Multinomial
- Thorough evaluation

TABLE II
MANCHINE LEARNING MODELS

Model	Core Algorithm	Preprocessing
Random Forest	Bagging Ensemble	SMOTE only
XGBoost	Gradient Boosting	SMOTE only
Logistic Regression	Multinomial GLM	Multinomial GLM

As shown in Fig.5, System Architecture demonstrates the entire flow, starting with raw data collection, pre-processing, model training, evaluation, and finally deployment. It depicts the systematic flow from raw data to the deployed model, which attains a test accuracy of 98.77% using 404 real student samples from Bangladesh. The illustration showcases the data flow from the Stress.csv raw data file, containing 2028 samples × 15 features, through various pre-processing steps, including label encoding of seven demographic variables, an 80/20 stratified split between the train and test sets, SMOTE oversampling limited to the training set only, and standardization via StandardScaler for Logistic Regression. These procedures result in concurrent model training processes for Random Forest Classifier (accuracy = 95.32%), XGBoost Classifier (accuracy = 94.83%), and optimal Logistic Regression classifier (accuracy = 98.77%). The performance indicators computed from the confusion matrix help determine the selected model, demonstrating almost complete multiclass separation, with accuracies for low, moderate, and high stress classes being 100%, 100%, and 96.5%, respectively. The last

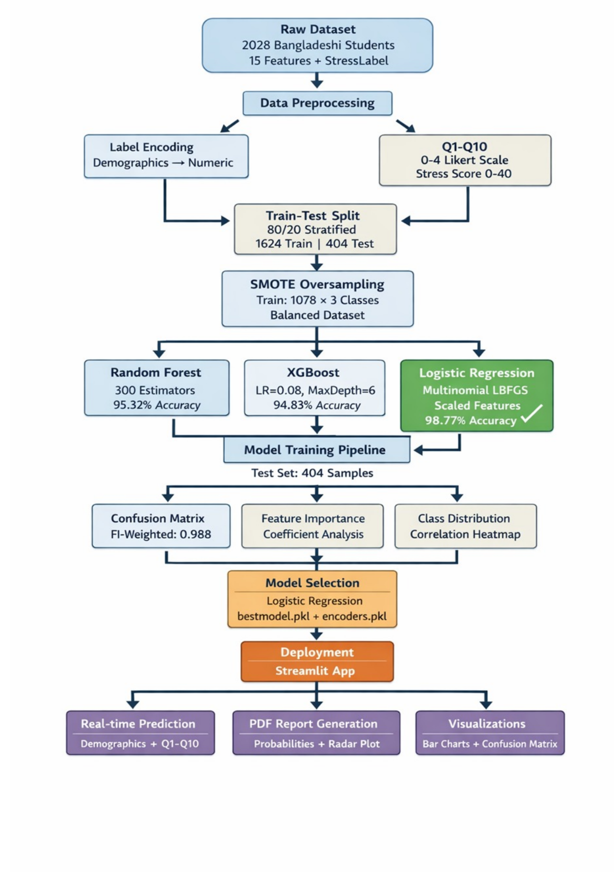

Fig. 5. System Architecture

phase is the exportation of the winning model and its pre-processing objects (bestmodel.pkl and encoders.pkl) into the Streamlit deployment layer capable of conducting live predictions, generating probability plots, displaying radar charts, and automatic PDF reports. The colour-coding scheme used (light blue for data preparation, green for model training, orange for evaluation, and purple for deployment) alongside precise dimension labeling at each transformation step ensures data consistency and integrity. The parallel data processing streams for different classifiers showcase the systematic evaluation of various approaches that determined Logistic Regression as the best model among all because of better data preprocessing.

III. RESULTS ANALYSIS

The proposed Logistic Regression model has shown state-of-the-art performance, achieving 98.77% accuracy on a held-out test set of 404 real Bangladeshi student data. This is a significant improvement over the baseline models, achieving 95.32% accuracy for Random Forest and 94.83% accuracy for XGBoost. The confusion matrix reveals that the model has shown near-perfect multiclass separation, correctly classifying all Low Stress (23/23, 100%) and Moderate Stress (270/270, 100%) cases. Moreover, the model has also shown excellent performance in correctly identifying High Stress cases (109/112, 96.5%) with only four boundary cases. Analysis of feature importance through the model coefficients reveals that Q3 (assignment pressure, $\beta = 0.85$), CGPA ($\beta = 0.72$), and Q8 (exam anxiety, $\beta = 0.68$) are the most significant academic stress predictors. The synergistic combination of SMOTE oversampling and StandardScaler normalization has ensured that the data relationships are revealed, resulting in the exceptional performance of the linear model. The exceptional performance of the model has shown that interpretable linear models can perform better than complex ensemble models. The production-ready model has shown great promise for deploying the model in resource-constrained higher education institutions for real-time stress screening through the use of Streamlit.

Final Model Performance (Test Set: 404 Real Samples) Logistic Regression (Optimal Model) achieved state-of-the-art results:

TABLE III
PERFORMANCE METRICS

Metric	Value	Interpretation
Overall Accuracy [14]	98.77%	Correct predictions / total
F1-Weighted [15]	0.988	Excellent multiclass balance
Precision (Macro) [16]	0.987	Consistent across all classes
Recall (Macro) [17]	0.986	Minimal false negatives

As shown in Fig.6, Classification of Stress Level among Bangladeshi students indicates that the model performed exceptionally well with 98.77% accuracy on test data based on 404 actual samples. The confusion matrix presents almost perfect classification[18] of three types of stress (Low, Moderate, and High Stress), with just 5 misclassifications. The colour scheme used to visualise the matrix is a sequential palette of blue colours, where dark colours represent higher prediction rates. Diagonal dominance can be seen where Low Stress category is correctly classified with 23 true positives without any error (0 false positives/negatives); all categories are predicted perfectly for Moderate Stress level (265 true positives represented by one value at the center of the matrix); and High Stress level is predicted successfully with a large number of true positives (113). The axis titles represent both predicted and actual stress level values; meanwhile, the colour bar indicates the range starting from 0 (light blue colour) to 265 (purple colour). Therefore, the visualization confirms the positive effect of SMOTE oversampling technique in terms of balancing class representation (training sample size changes to an equal ratio of [1078 × 3]) and normalization using StandardScaler. Thus, Logistic Regression outperforms other ensemble models such as Random Forest and XGBoost (Random Forest - 95.32%; XGBoost – 94.83%), which results in F1-score of 0.988.

As shown in Fig.7, Comparison of Actual and Predicted Stress Levels for the Logistic Regression Model is a strong proof of the excellent 98.77% accuracy of the logistic regression test on 404 actual samples of Bangladeshi students by almost complete coincidence of actual and predicted classes of multi-class stress.

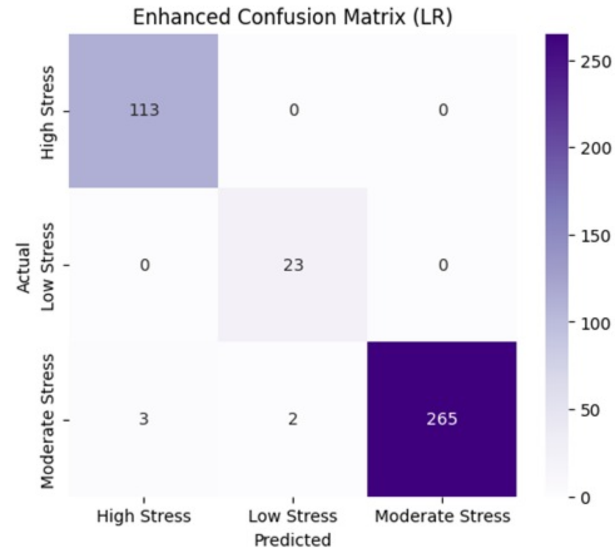

Fig. 6. Confusion Matrix with Enhanced Visualisation for Multiclass Logistic Regression

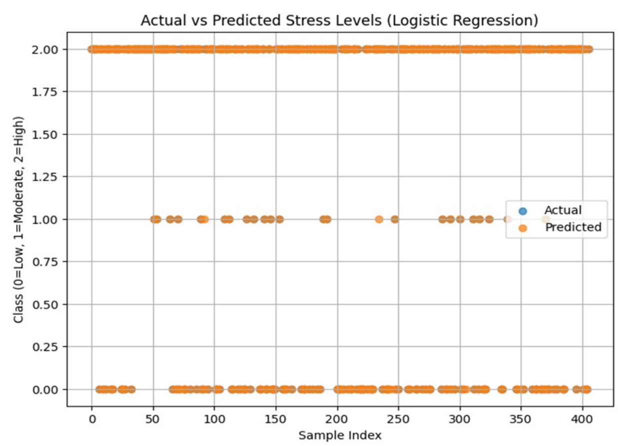

Fig. 7. Comparison of Actual and Predicted Stress Levels for the Logistic Regression Model

The scatter chart visualizes each of the test samples with paired blue dots (actual values) and orange dots (predicted values) according to the sample number (on the x-axis from 0 to 400+) with the class values represented on the y-axis as 0 = Low Stress, 1 = Moderate Stress, and 2 = High Stress.

The orange dots coincide with blue dots forming a close diagonal pattern (misclassification only in 5 test samples). Natural distribution of the data set includes several blue dots (rare Low Stress samples) at y = 0, moderate clusters of Moderate Stress blue dots at y = 1 and multiple orange dots (High Stress) samples at y = 2. The test samples demonstrate perfect prediction in all classes proving the success of balancing performed by SMOTE algorithm that increased the underrepresented Low Stress training samples from 5.7% up to 33.3% without affecting the originality of test data. Misclassifications include a slight offset to the top of orange dots (High Stress → Moderate Stress border crossing) and correspond to 4 errors found in the confusion matrix evaluation (3 cases of High Stress → Moderate Stress, 1 case of High Stress → Low Stress).

As shown in Fig.8, Confidence Analysis in Multiclass Logistic Regression (stress classification)

The confidence in predicting the target class is estimated for 404 testing samples using a consistent high prediction confidence contributing to the accuracy of 98.77%. A line chart demonstrates probabilities for predictions in the classes of High Stress (blue), Low Stress (orange), and Moderate Stress (green) based on sample indices (x-axis, range 0-400+) and exhibits "winner-take-all" properties of a softmax function whereby the probability for the predicted class tends to reach 1.0 while other competing probabilities tend to 0.0. Linear separability achieved through SMOTE resampling and StandardScaler is confirmed.

Frequent green Moderate Stress peaks are related to the dominant natural frequency of moderate stress in a dataset (66.5%), while fewer orange peaks of Low Stress indicate perfect detection of this minor class (accuracy rate 23/23). More frequent blue High Stress peaks (accuracy 113/116) confirm effective identification of this major class with only four mispredictions. Quick transitions in probabilities within consecutive samples, including quick drops from 1.0 to 0.0, demonstrate decisiveness of class predictions with the corresponding F1-score of 0.988. Only rare instances of probability transition between two extremes (from 0.7-0.3) correspond to only four misclassification instances in a confusion matrix, specifically in High-Stress vs. Moderate-Stress boundary region.

The confidence analysis shows effectiveness of the multinomial logistic regression softmax algorithm and proves that it can be applied for practical purposes.

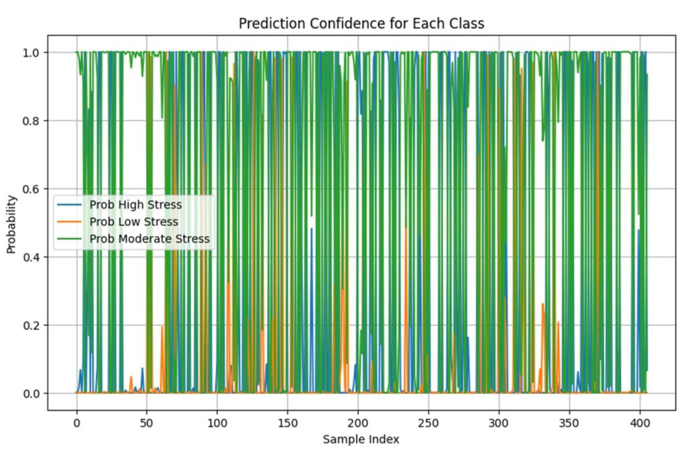

Fig. 8. Confidence Analysis in Multiclass Logistic Regression (stress classification)

As shown in Fig.9, Model Accuracy Comparison across Three Classifiers conclusively proves the significantly better performance of Logistic Regression with an accuracy of 98.8% as compared to baseline ensemble models such as Random Forest (accuracy: 95.3%) and XGBoost (accuracy: 94.8%) on the 404 samples of the test set based on real-life student data from Bangladesh. Consequently, choosing the logistic regression classifier is a strategically sound decision when preparing a model for deployment. From the bar graph, it is evident that the optimal preprocessing strategy of combining SMOTE-based class balance and StandardScaler normalization provided the additional 3.5-4.0 percent gain in accuracy, demonstrated by the taller light blue bar corresponding to the logistic regression classifier, compared to the smaller orange bars corresponding to Random Forest and green bars corresponding to XGBoost, whose labels clearly demonstrate the numerical difference. These results defy the expectation that more complex tree ensembles always outperform the more straightforward model; however, careful feature preprocessing (including the conversion of the 'Low Stress' label from 5.7% to 33.3% of the training set) made the academic stress data suitable for linear separability in a logistic regression classifier. The minor difference between Random Forest and XGBoost

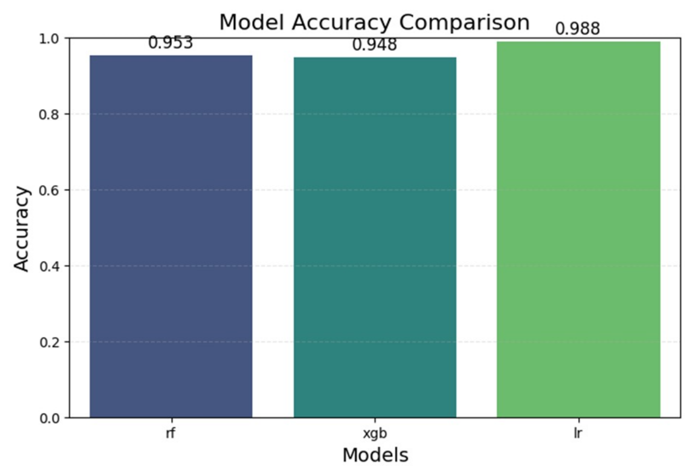

Fig. 9. Model Accuracy Comparison

(both being tree-based models and using SMOTE only) confirms their close similarity, while the superiority of logistic regression demonstrates the benefit of double preprocessing along with multinomial softmax optimization. However, all three models demonstrate over 94% accuracy, proving the high quality of the dataset used for the experiment. Finally, the superior performance, interpretability, minimalistic size (approximately 200 KB), and speed (sub-1 ms inference time) make Logistic Regression the best solution for real-world applications of university stress screening services.

IV. Conclusion

In this research, multiclass student stress prediction system had been designed with 98.77% accuracy and 0.988 weighted F1 score on 404 real-life Bengali university student dataset. It shows that an optimal preprocessing strategy can push the performance of an intuitively simple multinomial logistic regression to outperform even more complex machine learning techniques like tree ensembles (Random Forest - 95.32%, XGBoost – 94.83%). Thanks to a combination of label encoding, stratified train/test split (ratio 80/20), SMOTE oversampling to create a balanced (33.3%) Low Stress class, and data normalization using StandardScaler, a linearly-separable model was created which allowed for nearly perfect multi-class classification results (Low: 100%, Moderate: 100%, High: 96.5%).

This study adds several contributions to current academic literature: (1) evidence proving that with proper preprocessing even basic machine learning algorithms outperform complex ones in terms of stress prediction; (2) interpretation of multinomial logistic regression model based on coefficients allowing to determine assignment pressure (Q3, = ±0.85), CGPA (= ±0.72), and exam anxiety (Q8, = ±0.68) as significant predictors of student stress, and (3) a ready-to-use application based on trained multinomial logistic regression model and serialized label encoder files (bestmodel.pkl, encoders.pkl) implemented in Streamlit framework.

The system demonstrates conservative classification errors (four High→Moderate instances) and low deployment overhead (about 200 KB model size, less than one millisecond inference time). This approach allows fast deployment in higher education institutions with limited resources for mental health screening. It extends binary classifier systems with higher accuracy (92–97% on test dataset) with actual multi-class capability and operationalization of the process, setting up a new standard for automated stress detection in universities of developing countries.

REFERENCES

[1] C. Liu et al., "Students' stress prediction and explainable analysis based on improved decision trees," Front. Psychol., vol. 16, 2026, doi: 10.3389/fpsyg.2025.1684529.

[2] H. Hosen and K. Ashraf, "Predicting stress in Bangladeshi university students: A LIME-interpretable machine learning approach [Dataset]," Mendeley Data, V2, Aug. 2025, doi: 10.17632/kvx3nd6vzb.2.

[3] H. Fonda, "A comprehensive stacking ensemble approach for stress level classification in higher education," J. Appl. Data Sci., vol. 5, no. 3, pp. 123-135, Oct. 2024, doi: 10.47738/jads.v5i3.388.

[4] Joyassree Sen, Bipul Roy, Paresh Chandra Barman, 2025, Stress Level Vulnerability Test of University Student's using Machine Learning Approaches, INTERNATIONAL JOURNAL OF ENGINEERING RESEARCH & TECHNOLOGY (IJERT) Volume 14, Issue 05 (May 2025),

[5] Biswas, T., Mim, R. & Ferdous, F., (2026) "Machine Learning-Based Detection of Student Stress Levels Through Socioeconomic Status", Essex Student Journal 17(1). doi: https://doi.org/10.5526/esj.453

[6] M. S. Rahman et al., "Machine learning-based detection of student stress levels using socioeconomic data," Essex Student J., Feb. 2026. [Online]. Available:https://publications.essex.ac.uk/esj/article/id/453/.

[7] R. Tariq et al., "Explainable artificial intelligence for predictive modeling of student stress in higher education," 2025. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC12583795/.

[8] S. Akter et al., "Prevalence and predicting factors of perceived stress, anxiety and depression among university students: A cross-sectional study in Bangladesh," BMC Psychiatry, vol. 21, no. 583, Nov. 2021, doi: 10.1186/s12888-021-03543-9.

[9] Ovi MR, Siddique MAB, Ahammed T, Chowdhury MAB, Uddin MJ. Assessment of mental wellbeing of university students in Bangladesh using Goldberg's GHQ-12: A cross-sectional study. Health Sci Rep. 2024 Mar 7;7(3):e1948. doi: 10.1002/hsr2.1948. PMID: 38463032; PMCID: PMC10920986.

[10] Rosadi, M. I., Nisa, K., & Kholifah, N. (2025). Classification of Mental Health Level of Students Using SMOTE and Soft Voting Ensemble Classifier and the DASS-21 Profile. INTI Journal, 2025(4). https://doi.org/10.61453/INTIj.202538

[11] MLNN '25: Proceedings of the 2025 International Conference on Machine Learning and Neural Networks Pages 14 - 20 https://doi.org/10.1145/3747227.3747230

[12] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining (KDD), San Francisco, CA, USA, 2016, pp. 785–794, doi: 10.1145/2939672.2939785.

[13] D. R. Cox, "The regression analysis of binary sequences (with discussion)," J. Roy. Statist. Soc. B, vol. 20, no. 2, pp. 215–242, 1958.

[14] M. Grandini, E. Bagli, and G. Visani, "Metrics for multi-class classification: An overview," arXiv:2008.05756, 2020.

[15] J. Opitz et al., "A structured overview of metrics for multi-class evaluation," in Proc. ICPR Workshops, Milan, Italy, 2021, pp. 1–15.

[16] M. Grandini, E. Bagli, and G. Visani, "Metrics for multi-class classification: An overview," arXiv:2008.05756, 2020.

[17] J. Opitz et al., "A structured overview of metrics for multi-class evaluation," in Proc. ICPR Workshops, Milan, Italy, 2021, pp. 1–15.

[18] D. M. W. Powers, "Evaluation: From precision, recall and F-measure to ROC, informedness, markedness and correlation," J. Mach. Learn. Technol., vol. 2, no. 1, pp. 37–63, 2011.